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# T Cell-Mediated, IFN- $\gamma$ -Facilitated Rejection of Murine B16 Melanomas<sup>1</sup>

Waltraud Böhm,\* Stefan Thoma,\* Frank Leithäuser,<sup>‡</sup> Peter Möller,<sup>‡</sup> Reinhold Schirmbeck,\* and Jörg Reimann<sup>2\*</sup>

The murine melanoma cell line B16.F10 (H-2<sup>b</sup>) was used to study specific T cell responses that reject tumors. Stable B16 transfectants were established that express viral Ags, either the hepatitis B surface Ag (HBsAg) or the large tumor Ag (T-Ag) of SV40. B16 cells and their transfected sublines were CD40<sup>+</sup>CD44<sup>+</sup> but expressed no (or low levels of the) costimulator molecules CD154 (CD40L), CD48, CD54, CD80, and CD86. Surface expression of MHC class I (K<sup>b</sup>, D<sup>b</sup>) and class II (I-A<sup>b</sup>) molecules by B16 cells was low, but strikingly up-regulated by IFN- $\gamma$ . CD95 (Fas) and CD95 ligand (CD95L (FasL)) were “spontaneously” expressed by B16 cells growing in vitro in serum-free medium; these markers were strikingly up-regulated by IFN- $\gamma$ . B16 cells coexpressing CD95 and CD95L were irreversibly programmed for apoptosis. In vitro, noninduced B16 transfectants stimulated a specific IFN- $\gamma$  release response, but no cytolytic response (in a 4-h assay) in MHC class I-restricted CTL; in contrast, IFN- $\gamma$ -induced B16 targets were efficiently and specifically lysed by CTL. In vivo, B16 transfectants were specifically rejected by DNA-vaccinated syngeneic hosts through a T-dependent immune effector mechanism. The tumors showed evidence of massive apoptosis in vivo during the rejection process. The data suggest that CTL-derived IFN- $\gamma$  enhances an intrinsic suicide mechanism of these tumor cells in addition to facilitating lytic interactions of effectors with tumor targets. *The Journal of Immunology*, 1998, 161: 897–908.

**T**ransplantable murine melanomas are well-established models for the study of experimental cancer therapies. Melanomas express different tumor-associated Ags, which are potential targets for novel designs of therapeutic cancer vaccines. Melanoma-associated Ags include tyrosinase and tyrosine-related protein (TRP)-1/2 (1–3) or the MAGE and BAGE class of Ags (4–9). Many immunotherapeutic protocols have been tested using the murine B16 melanoma cell line (and its sublines) that originated in the C57BL/6 (H-2<sup>b</sup>) mouse strain. B16 melanoma cells express some of these melanoma-associated Ags, and their growth can be controlled in vivo by different classes of specific or nonspecific immune effector cells.

Some tumor cells express the CD95 receptor for the TNF-like CD95 ligand (CD95L).<sup>3</sup> These tumors can potentially be eradicated by CD95L produced locally by tumor-infiltrating lymphomyeloid cells (10). Cells from primary or metastatic lesions of some tumors, and from some in vitro established tumor cell lines, express the CD95L; this was shown for colonic adenocarcinomas (11), hepatocellular carcinomas (12, 13), leukemias (14), and melanomas (15). These CD95L-expressing tumor cells can escape immune surveillance by inducing apoptosis of CD95<sup>+</sup> immune effector cells (16, 17). Coexpression of CD95 and CD95L by normal

and tumor cells has been found. Cells of some activated lymphocyte subsets coexpress CD95 and CD95L. These cells undergo apoptosis (or “activation-induced cell death”), and this mechanism seems to represent one form of a physiologic down-regulation of an immune response (18–20). Although CD95 is abundantly expressed by hepatocytes, CD95L<sup>+</sup> hepatocellular carcinoma cells down-regulate Fas expression to escape apoptosis (12). Colonic adenocarcinoma cells or leukemia and lymphoma cells coexpress CD95 and CD95L but acquire resistance to CD95-induced apoptosis (11, 14).

Although expression of MHC-I/II and costimulator molecules on the surface of noninduced B16 tumor cells is low or absent, primed CTL can play a role in rejecting this tumor. In vitro, presentation of antigenic peptides in the context of MHC-I glycoproteins by B16.F10 cells to CD8<sup>+</sup> CTL lines (CTL<sup>L</sup>) triggers IFN- $\gamma$  (IFN- $\gamma$ ) release but not specific lysis of the tumor cells by CTL. IFN- $\gamma$  enhances MHC-I expression by B16.F10 cells and thereby renders them susceptible to CTL-mediated specific lysis. In addition, we describe the cytokine-amplified apoptosis of murine B16.F10 melanoma cells. By enhancing expression of CD95 and CD95L by a subset of B16 cells, CTL-derived IFN- $\gamma$  irreversibly commits this tumor cell subset to apoptosis. In vivo, T cells efficiently reject B16.F10 tumor cells. This suggests that IFN- $\gamma$ -dependent apoptosis of CD95L<sup>+</sup>/CD95<sup>+</sup> tumor cells can play a role in tumor rejection in vivo.

## Materials and Methods

### Mice

C57BL/6J mice (H-2<sup>b</sup>) were bred and kept under specific pathogen-free conditions in the animal colony of the University of Ulm. Breeding pairs of these mice were obtained from Bomholtgard (Ry, Denmark). Male and female mice were used at 12 to 16 wk of age.

### Cell lines

The Rauscher virus-transformed T lymphoma line, RBL5, is derived from a C57BL/6J (H-2<sup>b</sup>) mouse. The RBL5/S and RBL5/T transfectants (expressing the hepatitis B surface Ag (HBsAg) or the SV40 large tumor Ag (T-Ag)) have been described (21–23). The H-2<sup>b</sup> melanoma cell lines

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<sup>3</sup> Abbreviations used in this paper: L, ligand (e.g., CD95L, CD40L, FasL); MHC-I/II, MHC class I/II; T-Ag, large tumor Ag of SV40; HBsAg, hepatitis B surface Ag; FCM, flow cytometry; CTL<sup>L</sup>, CTL line; PE, phycoerythrin; TdT, terminal deoxynucleotidyl transferase; TUNEL, TdT-mediated dUTP-biotin nick end labeling; 7-AAD, 7-amino-actinomycin D.

B16.F0 and B16.F1 from the American Type Culture Collection (ATCC, Manassas, VA; CRL-6322, CRL-6323) and B16.F10 (from Dr. P. Antonsson, Lund, Sweden) were used. In most experiments, we used the B16.F10 line.

The bovine papilloma virus-based vector BMGneo (24) (a generous gift from Drs. Y. Karasuyama and F. Melchers, Basel, Switzerland) was used to construct the BMG/HBS expression plasmid (22) and the BMG/T-Ag.1 expression plasmid (21). BMG/HBS and BMG/T-Ag.1 vector DNA, or BMGneo vector DNA (without insert), were transfected into B16 cells using the CaPO<sub>4</sub> method. Expression of the viral proteins was tested in B16 transfectants labeled with 400  $\mu$ Ci of [<sup>35</sup>S]methionine (cat. no. SJ1015; Amersham, Braunschweig, Germany) in methionine-free RPMI 1640 medium. Labeled cells were washed twice in PBS and extracted with lysis buffer (120 mM NaCl, 1% aprotinin (Trasylol, cat. no. 48764; Bayer, Leverkusen, Germany), 50  $\mu$ M leupeptin, 0.5% Nonidet P-40, 10% glycerol, and 50 mM Tris/HCl, pH 8.0) for 30 min at 4°C. Extracts cleared by centrifugation (30 min, 20,000  $\times$  g, 4°C) were subsequently incubated for 2 h at 4°C with protein-A Sepharose (cat. no. 17-0780-01; Pharmacia, Freiburg, Germany). The HBsAg protein was precipitated with a polyclonal rabbit anti-HBs antiserum and protein A-Sepharose, and the T-Ag protein was precipitated with the mAb Pab108 (25) and protein A-Sepharose at 4°C. The precipitates were washed extensively in 0.5 M LiCl, 1% Nonidet P-40, and 0.1 M Tris/HCl (pH 9.0). After two additional washes of the complexes in 1 $\times$  PBS and one wash in 0.1 $\times$  PBS, they were recovered from protein-A Sepharose in 400  $\mu$ l of elution buffer (1.5% SDS, 5%  $\beta$ -ME, and 7 mM Tris/HCl, pH 6.8). Following an incubation for 30 min at 37°C, SDS-denatured eluates were lyophilized and dissolved in 30  $\mu$ l of aqueous solution of 7%  $\beta$ -ME, 10% glycerol and bromophenol blue. After boiling for 2 min, samples were analyzed by SDS-PAGE.

#### IFN- $\gamma$ treatment of B16 cells

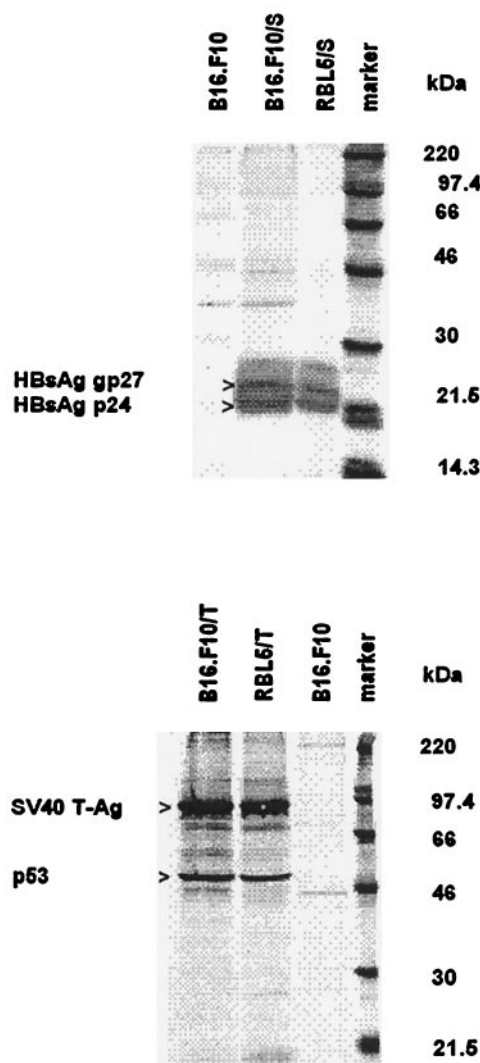
B16 cells were cultured for 12 to 48 h in serum-free medium containing 20 to 400 U/ml mouse rIFN- $\gamma$  (cat. no. 1276905, Boehringer Mannheim, Mannheim, Germany). Cells were harvested and washed, then used in flow cytometry (FCM) analyses or cytotoxic assays.

#### FCM analyses

For FCM studies, cells were suspended in PBS/0.3% (w/v) BSA supplemented with 0.1% (w/v) sodium azide. Nonspecific binding of Abs to FcR was always blocked by preincubating cells with the mAb 2.4G2 (cat. no. 01241D) directed against the Fc $\gamma$ III/II (CD16/CD32) (2  $\mu$ g mAb/10<sup>6</sup> cells/100  $\mu$ l). Cells were incubated with 0.5  $\mu$ g/10<sup>6</sup> cells of the relevant mAb for 20 min at 4°C and washed twice. In most experiments, cells were subsequently incubated with a second-step reagent for 20 min at 4°C. Three-color FCM analyses were performed on a FACScan (Becton Dickinson, Mountain View, CA). The forward narrow angle light scatter was used as an additional parameter to facilitate exclusion of dead cells and aggregated cell clumps. The following reagents and mAbs from PharMingen (Hamburg, Germany) were used: FITC-conjugated or biotinylated anti-CD3 $\epsilon$  mAb 145-2C11 (cat. no. 01084D, 01082D), FITC-conjugated anti-CD4 (L3T4) mAb H129.19 (cat. no. 09004D), biotinylated anti-CD4 (L3T4) mAb RM4-5 (cat. no. 01062D), anti-CD8 $\alpha$  (Ly-2) mAb 53-6.7 (cat. no. 01042D), FITC-conjugated anti-CD54 mAb 3E2 (cat. no. 01224D), FITC-conjugated anti-CD44 (Pgp-1) mAb IM7 (cat. no. 01224D), phycoerythrin (PE)-conjugated anti-CD80 (B7.1) mAb 16-10A1 (cat. no. 0965B), biotinylated anti-CD86 mAb GL-1 (cat. no. 09272D), biotinylated anti-CD95 (Fas) mAb Jo2 (cat. no. 15402D), FITC-conjugated anti-H2-K<sup>b</sup> mAb AF6-88.5 (cat. no. 06104D), biotinylated anti-H2-D<sup>b</sup> mAb 28-14-8 (cat. no. 06232D), FITC-conjugated anti-IA<sup>b</sup> mAb 25-9-17 (cat. no. 06254D), PE-conjugated anti-CD48 mAb HM48-1 (cat. no. 09175B), biotinylated anti-CD40 mAb 3/23 (cat. no. 09662D), biotinylated anti-CD40L (gp39) mAb MR-1 (cat. no. 09022D), Via-Probe (7-AAD; cat. no. 34321X), Annexin V-PE-conjugated (cat. no. 65875X), Annexin V-FITC-conjugated (cat. no. 65874X), and PE-, cychrome-, and FITC-conjugated streptavidin (cat. no. 13025D, 13024D). The anti-CD95L (FasL) mAb N-20 (cat. no. sc-834) was obtained from Santa Cruz Biotechnology (Heidelberg, Germany).

#### HBsAg- and T-Ag-encoding expression vectors used for DNA immunization

The construction of the expression plasmid pCMV-1/T has been described (26). The expression plasmid pCI/S (containing the small HBsAg) was constructed by cloning the HBsAg-encoding Xho-I/Bgl-II fragment of HBV, subtype ayw (from the plasmid pTKTHBV2, a generous gift of Dr. M. Meyer, Munich, Germany) into the pCI vector (cat. no. E1731, Pro-

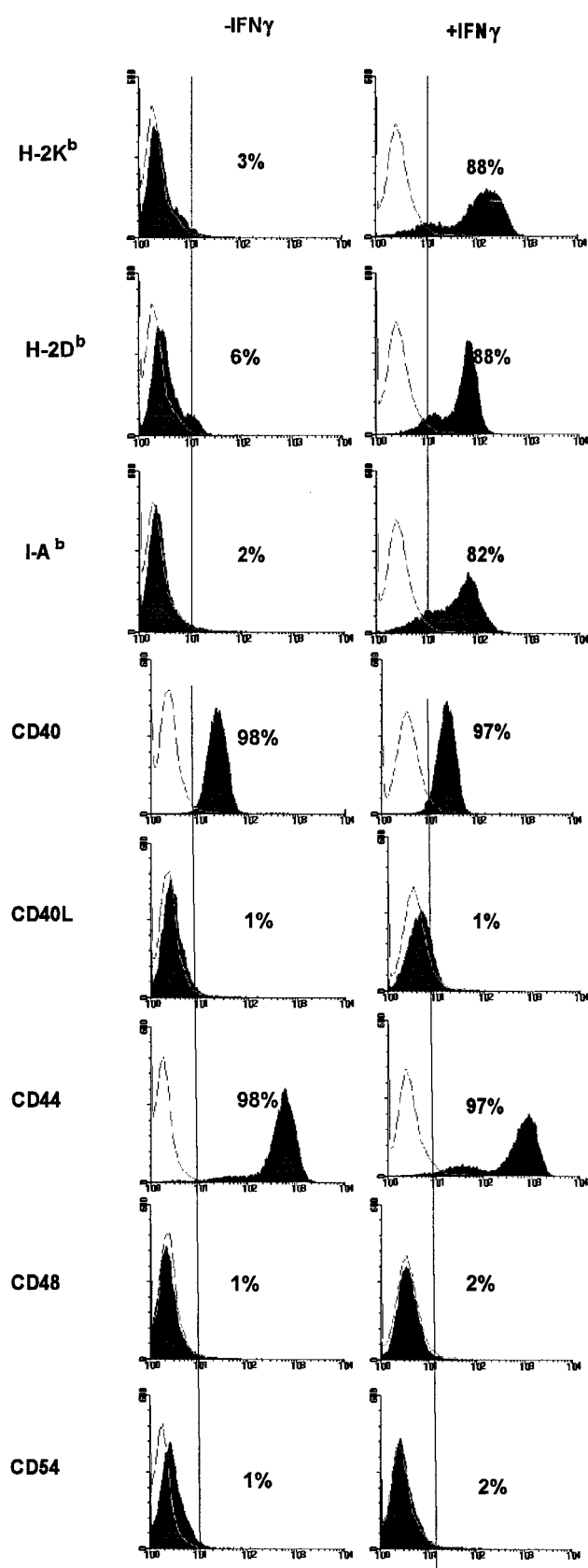


**FIGURE 1.** Expression of the viral proteins in B16.F10 melanoma cells. B16.F10 melanoma cells were transfected with the (HBsAg-encoding) BMG/HBS plasmid DNA, or the (T-Ag-encoding) BMG/T-Ag.1 plasmid DNA, or BMGneo vector DNA (containing no insert). Expression of the viral proteins was detected in cells (selected for 8 wk) by immunoprecipitating the Ag from cell lysates using specific Abs (see *Materials and Methods* for details). Control immunoprecipitations from lysates of RBL5/S and RBL5/T transfectants are included into the SDS-PAGE analysis.

mega, Heidelberg, Germany) cut with Xho-I/Bam-HI. COS-7 cells (CRL-1651, ATCC) were transfected with 30  $\mu$ g plasmid DNA/10<sup>6</sup> cells using the CaPO<sub>4</sub>-method. Cells were cultured for 2 d, and expression of the viral proteins by these cells was detected by immunoprecipitation as described above. HBsAg and T-Ag proteins were expressed in transiently transfected COS-7 cells (data not shown). Plasmid DNA used for immunization was purified by anion exchange chromatography using the Qiagen maxiprep kit (Qiagen, Hilden, Germany). Naked plasmid DNA suspended at 1  $\mu$ g/ $\mu$ l in PBS was injected.

#### Reverse transcriptase-PCR

Total RNA was isolated from nontransfected or transfected B16 cells (Qiagen minikit, cat. no. 14123) and reverse transcribed to cDNA using a cDNA synthesis kit (Stratagene, Frankfurt, Germany cat. no. 200420). Aliquots from each cDNA preparation were PCR-amplified using the forward/reverse primers and the conditions previously described (27), and the buffer and polymerase from Stratagene (cat. no. 600153-81, 600153-82). The 0.54-kb fragment was resolved on a 2% agarose gel.



**FIGURE 2.** Surface expression of costimulating molecules by B16.F10 cells. B16.F10 cells cultured in serum-free medium were unstimulated ( $-IFN\gamma$ ) or stimulated for 12 h with 50 U/ml  $IFN\gamma$  ( $+IFN\gamma$ ). Surface expression of the MHC-I molecules K<sup>b</sup> and D<sup>b</sup>, of the MHC-II molecules I-A<sup>b</sup>, and of the costimulator molecules CD40, CD40L, CD44, CD48, and CD54 was measured using the mAb listed in the *Materials and Methods*.

### DNA vaccination

We injected 50  $\mu$ l PBS containing 50  $\mu$ g plasmid DNA into each regenerating tibialis anterior muscle 5 days after the injection of cardiotoxin (Latoxan; Rosans, France) as described (28). All mice received one bilateral intramuscular injection. Mice injected with pCMV-1/T plasmid DNA or pCI/S plasmid DNA developed a potent T-Ag- or HBsAg-specific CTL response (23, 26, 28). Noninjected mice or mice injected with pCI or pCMV-1 plasmid DNA (without insert) were used as controls.

### Subcutaneous transplantation of tumor cells

B16.F10, B16.F10/S, and B16.F10/T cells were cultured in serum-free medium (UltraCulture; BioWhittaker, Walkersville, MD, cat. no. 12-725F). The cells were transplanted into immunocompetent, syngeneic C57BL/6J mice by injecting titrated numbers of cells ( $10^3$ – $10^6$  cells/mouse) in 50  $\mu$ l PBS s.c. into the left lateral flank of age- and sex-matched mice. Tumor growth was measured every second or third day. Mice bearing tumors with a diameter  $>1$  cm were sacrificed (according to the regulation for animal experimentations of Baden-Württemberg). Data were plotted and statistically evaluated using the GraphPad Prism version 2 software (see Tables I and II).

Spleen cells were obtained from C57BL/6 mice primed 3 weeks before to T-Ag by DNA vaccination. Into B16.F10/T tumor-bearing mice, we injected i.p.  $2 \times 10^7$ -primed splenic (B cell-depleted) T cells. This led to an easily measurable regression of the 3 to 6-mm tumor in most treated mice. Regressing tumors were removed for histopathologic examination 2 to 3 days after T cell transfer.

### In vivo suppression of CD4<sup>+</sup> or CD8<sup>+</sup> T cells in mice

CD4<sup>+</sup> or CD8<sup>+</sup> T cell subsets were suppressed in mice by repeated injections of the anti-CD4 mAb YTS 191.1 or the anti-CD8 mAb YTS 169.4. Two days before the tumor cell transplantation, mice were i.p. injected with 200  $\mu$ l PBS containing 500  $\mu$ g purified Ab. At 5-day intervals (i.e., day 3 and day 8 after tumor cell transplantation) mice were again injected with 250  $\mu$ g of the respective Abs. FCM analyses of PBMC populations demonstrated that 90 to 98% of T cells expressing the respective phenotype were deleted in treated mice (data not shown).

### CTL lines

Spleen or lymph node cells were suspended in  $\alpha$ -MEM (Life Technologies, Berlin, Germany) supplemented with 10 mM HEPES buffer,  $5 \times 10^{-5}$  M 2-ME, antibiotics, and 10% v/v FCS (Pan Systems, Aidenbach, Germany). To the culture medium was further added 2% v/v of a selected batch of Con A-stimulated rat spleen cell supernatant. In upright 25-cm<sup>2</sup> tissue culture flasks in a humidified atmosphere/5% CO<sub>2</sub> at 37°C,  $3 \times 10^7$  responder cells were cocultured with  $1.5 \times 10^6$  syngeneic RBL5/S or RBL5/T transfectants (irradiated with 20,000 rad) in 10 ml of medium. CD3<sup>+</sup>CD8<sup>+</sup> T blasts harvested from restimulated cultures were separated on a discontinuous density gradient and restimulated twice weekly (at  $5 \times 10^4$  cells/ml) with irradiated RBL5/S or RBL5/T transfectants. These CTL lines (CTL) were grown for 2 to 6 wk in vitro. Cells from all lines were cytolytic and displayed the CD3<sup>+</sup>CD4<sup>+</sup>CD8<sup>+</sup>TCR $\alpha\beta$ <sup>+</sup> phenotype.

Medium was conditioned by coculturing  $10^6$ /ml CTL.L.T cells (or CTL.L.S cells) with  $10^6$ /ml B16.F10.T cells (or B16.F10.S cells) for 12 to 24 h at 37°C. The medium was cleared after the incubation (15 min, 2000  $\times$  g) and stored at 4°C. In the experiments described, the medium was used at 5% v/v. The mAb R4-6A2 neutralizes mouse  $IFN\gamma$  bioactivity; it was added to cultures to obtain a final concentration of 20  $\mu$ g/ml. In some experiments, the  $IFN\gamma$  bioactivity in conditioned medium was neutralized by incubating it with 20  $\mu$ g/ml of mAb R46A2 for 12 h at 4°C.

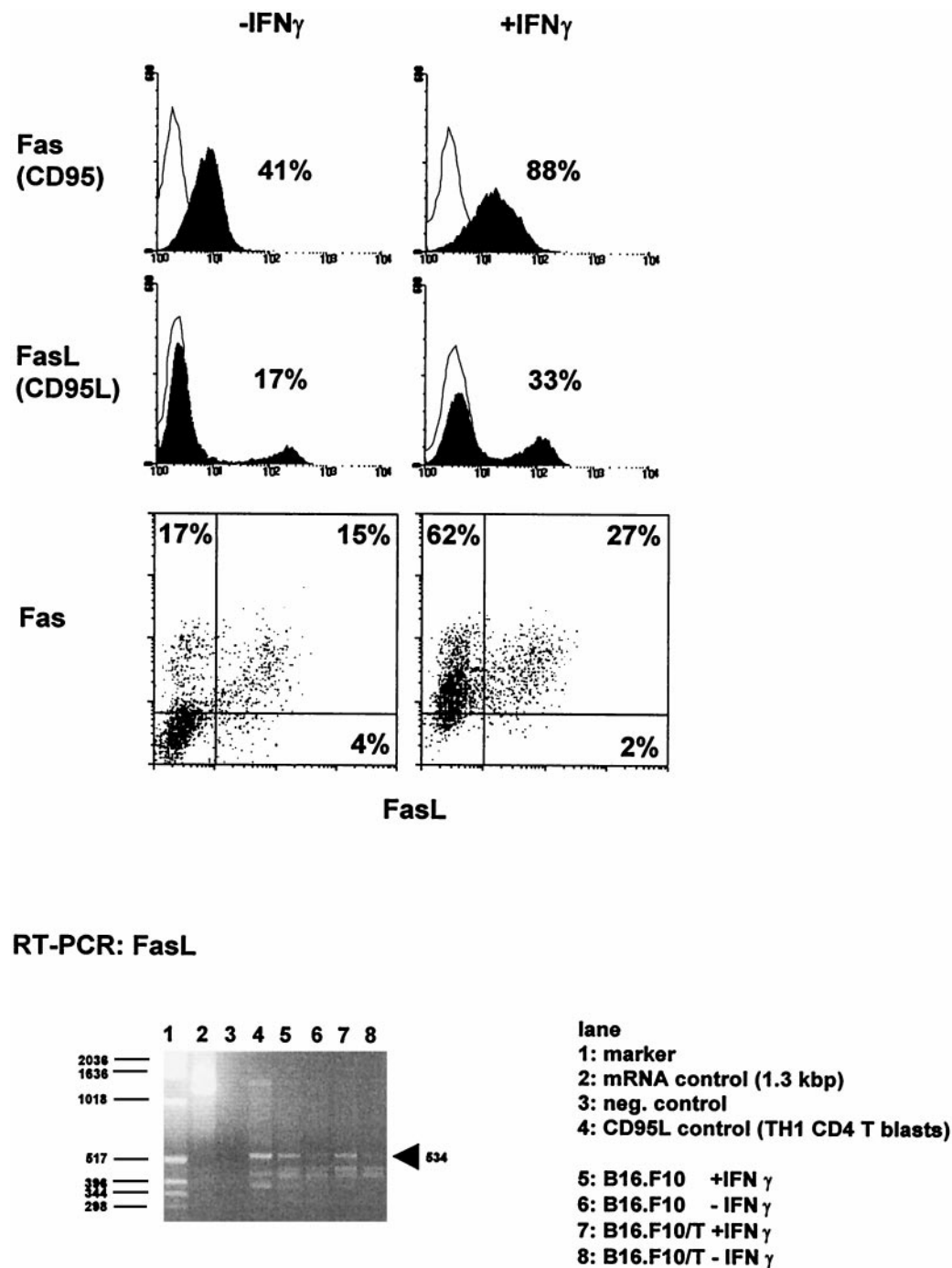
### In vitro culture and cytotoxic assay

CTL harvested after 5 days of in vitro culture were washed, and serial dilutions of these effector cells were cocultured with  $2 \times 10^5$  <sup>51</sup>Cr-labeled targets in 200- $\mu$ l round-bottom wells. After a 4-h incubation at 37°C, 100  $\mu$ l of supernatant were collected for gamma counting. The percentage of specific release was calculated as [(experimental release – spontaneous release)/(total release – spontaneous release)]  $\times$  100. Total counts were measured by resuspending target cells. Spontaneously released counts were always  $<15\%$  of the total counts. Data shown are the mean of triplicate cultures. The SEM of triplicate data was always  $<15\%$  of the mean.

### Histology studies

Cells undergoing apoptosis were detected in situ by labeling DNA strand breaks using terminal deoxynucleotidyl transferase (TdT)-mediated dUTP-biotin nick end labeling (TUNEL). Briefly, formalin-fixed and paraffin-embedded tissue sections were dewaxed and digested with proteinase K





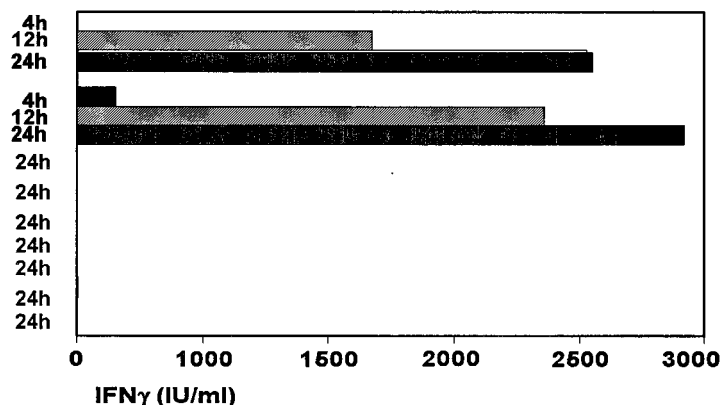
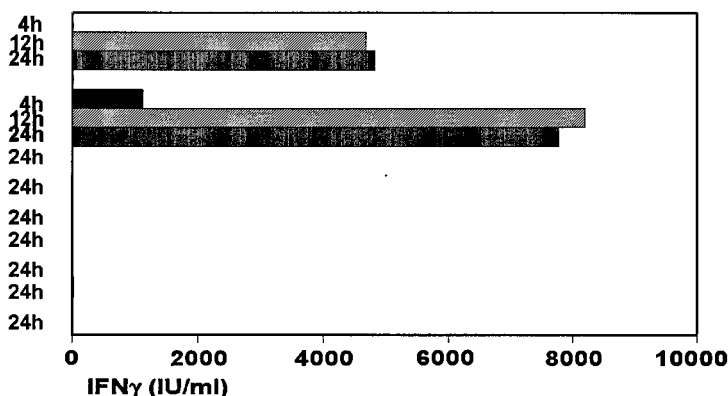
**FIGURE 3.** Surface expression of CD95 and CD95L by B16.F10 cells. B16.F10 cells growing in serum-free medium were unstimulated ( $-IFN-\gamma$ ) or stimulated for 12 h with 50 U/ml  $IFN-\gamma$  ( $+IFN-\gamma$ ). Surface expression and coexpression of CD95 (Fas)/CD95L (FasL) was determined using the mAb listed in *Materials and Methods*. In RT-PCR analyses of unstimulated or stimulated B16.F10 cells, mRNA of CD95L (FasL) was evident.

(20  $\mu$ g/ml). After inactivating endogenous peroxidase with  $H_2O_2$ , the labeling reaction was conducted using 10 U TdT (Promega) and 2  $\mu$ M digoxigenin-11-dUTP (Boehringer Mannheim) in 50  $\mu$ l TdT-buffer (0.5 M cacodylic acid, sodium salt, pH 6.8; 1 mM  $CoCl_2$ , 0.5 mM DTT; 0.05% (w/v) BSA; 0.15 M NaCl). Labeled cells were detected using sheep anti-digoxigenin Fab (5  $\mu$ g/ml; Boehringer-Mannheim) followed by horseradish peroxidase-conjugated F(ab')<sub>2</sub> fragment of donkey anti-sheep IgG (0.2 mg/ml; Dianova, Hamburg, Germany). Bound horseradish peroxidase was visualized by the substrate 3-amino-9-ethylcarbazole (0.1 mg/ml in 0.17 M sodium acetate, pH 5.2, plus 0.01%  $H_2O_2$ ). After counterstaining in Mayer's hematoxylin, representative tissue areas were photographed under a Zeiss Axiophot microscope.

### Results

#### *Expression of the viral proteins HBsAg and SV40 T-Ag in B16 melanoma cells*

B16.F10 cells were transfected with the episomal expression constructs BMG/HBS or BMG/T-Ag.1. Control transfectants were transfected with BMGneo vector DNA without insert. Transfected cells were selected for 5 to 10 wk in vitro. Immunoprecipitation studies revealed stable expression of the nonglycosylated p24 and the glycosylated gp27 form of the HBsAg protein by B16.F10/S

**responder stimulator****CTLL.T B16.F10/T****CTLL.T RBL5/T****CTLL.T B16.F10****CTLL.T B16.F10/T +  $\alpha$  IFN $\gamma$** **CTLL.T RBL5/T +  $\alpha$  IFN $\gamma$** **CTLL.T RBL5****CTLL.T -****- B16.F10/T****- RBL5/T****CTLL.S B16.F10/S****CTLL.S RBL5/S****CTLL.S B16.F10****CTLL.S B16.F10/S +  $\alpha$  IFN $\gamma$** **CTLL.S RBL5/S +  $\alpha$  IFN $\gamma$** **CTLL.S RBL5****CTLL.S -****- B16.F10/S****- RBL5/S**

**FIGURE 4.** IFN- $\gamma$  release by CTLL in response to stimulation by B16.F10 tumor cell transfectants. Representative data using two MHC-I-restricted CTLL specific for either HBsAg (CTLL.S) or SV40 T-Ag (CTLL.T) are shown that were generated in C57BL/6 mice by DNA vaccination. CTLL were stimulated for 4, 12, or 24 h with transfected B16.F10 or (positive control) transfected RBL5 cells. Controls included CTLL stimulated with nontransfected B16.F10 or RBL5 cells, transfected RBL5 cells, or B16.F10 cells cultured without CTLL, and supernatant in which IFN- $\gamma$  bioactivity was neutralized by the mAb R4-6A2 (anti-IFN- $\gamma$ ). IFN- $\gamma$  in conditioned medium was measured by ELISA.

transfectants, or of the T-Ag protein (complexed to p53) by the B16.F10/T transfectants (Fig. 1). Cells from these lines, and from sublines derived from these lines, showed stable expression of the viral proteins for longer than 8 mo. The level of expression of HBsAg and T-Ag by B16.F10/S or B16.F10/T transfectants was comparable to that of the RBL5/S and RBL5/T transfectants (Fig. 1) described previously (21, 22).

#### Surface phenotype of noninduced and IFN- $\gamma$ -induced B16 cells

B16 melanoma cell lines (derived from H-2<sup>b</sup> C57BL/6 mice) cultured in serum-supplemented or serum-free medium expressed only very low levels of the MHC-I molecules K<sup>b</sup> and D<sup>b</sup> and the MHC-II molecule I-A<sup>b</sup> on the cell surface. Expression of K<sup>b</sup>, D<sup>b</sup>, and I-A<sup>b</sup> was strikingly up-regulated in cells stimulated with IFN- $\gamma$  (Fig. 2). Similar data were obtained with nontransfected B16.F10 cells and the transfected B16.F10/S and B16.F10/T sublines (data not shown).

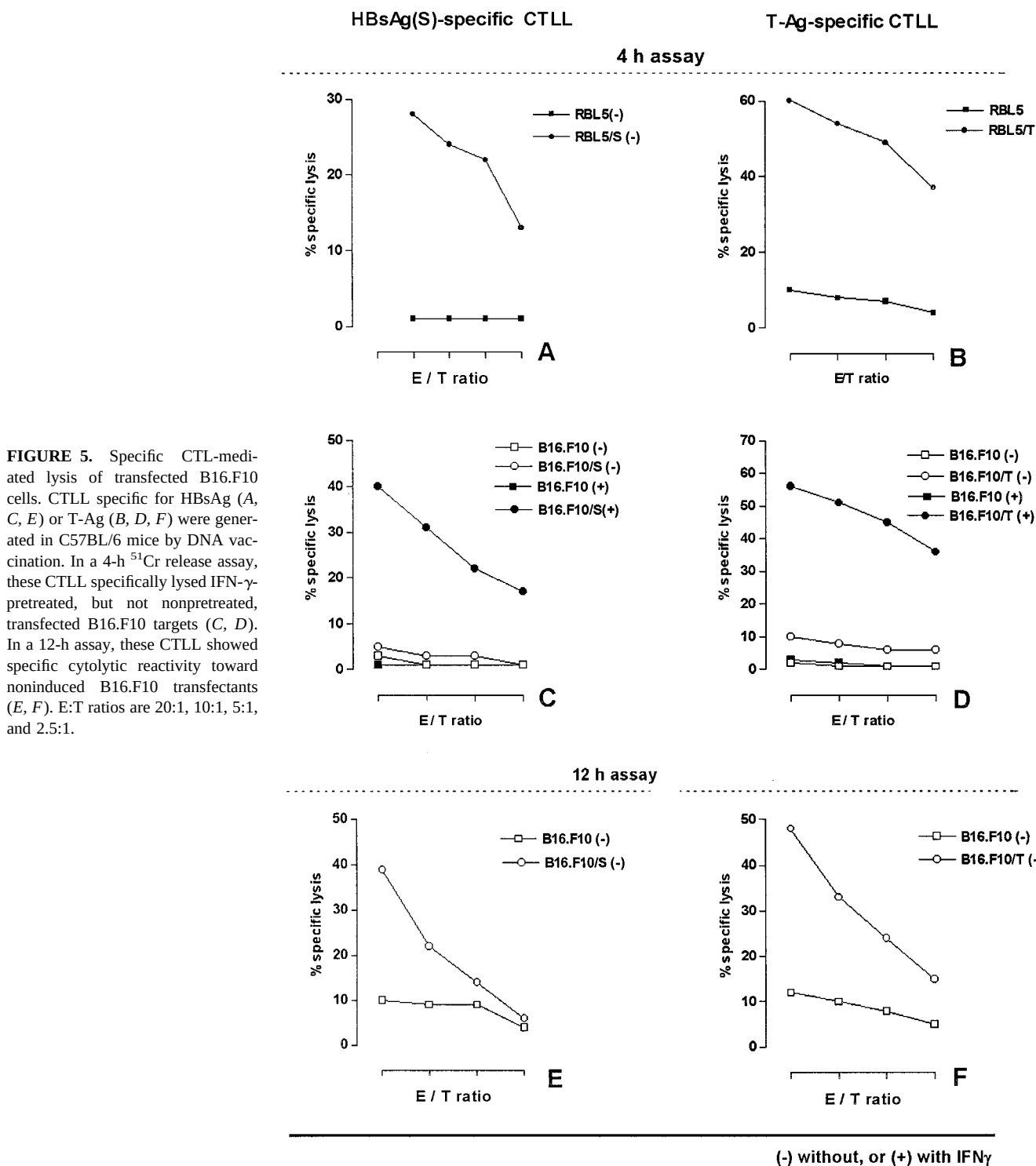
We analyzed the surface expression of T cell-costimulating molecules in noninduced and IFN- $\gamma$ -induced (nontransfected or transfected) B16 cells cultured in either serum-supplemented or serum-free medium. Nonpretreated and IFN- $\gamma$ -treated B16 cells express CD40 molecules of the NGFR superfamily on the cell surface, but no (or very low levels of) CD40L (Fig. 2). High surface expression of CD44 was detectable in noninduced and IFN- $\gamma$ -induced

B16 cells (Fig. 2). Expression of the costimulator molecules CD48 (interacting with murine CD2), CD54/ICAM-1 (interacting with LFA-1), CD80 (B7-1), and CD86 (B7-2) (interacting with CD28/CTLA4) on the surface of nonpretreated and IFN- $\gamma$ -treated B16 cells was low or absent (Fig. 2, and data not shown). The B16 cells, therefore, expressed the CD40<sup>+</sup>CD40L<sup>-</sup>CD44<sup>+</sup>CD48<sup>-</sup>CD54<sup>-</sup>CD80<sup>-</sup>CD86<sup>-</sup> phenotype. Similar data were found by analyzing B16.F10 melanoma cells and their transfected sublines cultured in serum-supplemented or serum-free medium.

#### Inducible surface expression of CD95 (Fas) and CD95L (FasL) in B16 cells

In vitro-cultured B16 cells express CD95 (Fas) and CD95L (FasL) on the cell surface (Fig. 3). The fractions of CD95<sup>+</sup> and CD95L<sup>+</sup> B16 cells were always higher when cells were cultured in serum-supplemented than in serum-free medium. IFN- $\gamma$  up-regulated CD95 and CD95L surface expression on B16 cells (Fig. 3). Expression of CD95L by noninduced and IFN- $\gamma$ -induced B16 cells was confirmed at the RNA level by RT-PCR analyses (Fig. 3). Two-color FCM analyses showed that a fraction of CD95<sup>+</sup> B16 cells coexpressed CD95L, while another fraction of B16 cells displayed the CD95<sup>+</sup>CD95L<sup>-</sup> surface phenotype (Fig. 3).

In IFN- $\gamma$ -stimulated cultures of B16 cells, the fraction of CD95<sup>+</sup>CD95L<sup>+</sup> B16 cells was increased two- to threefold. This



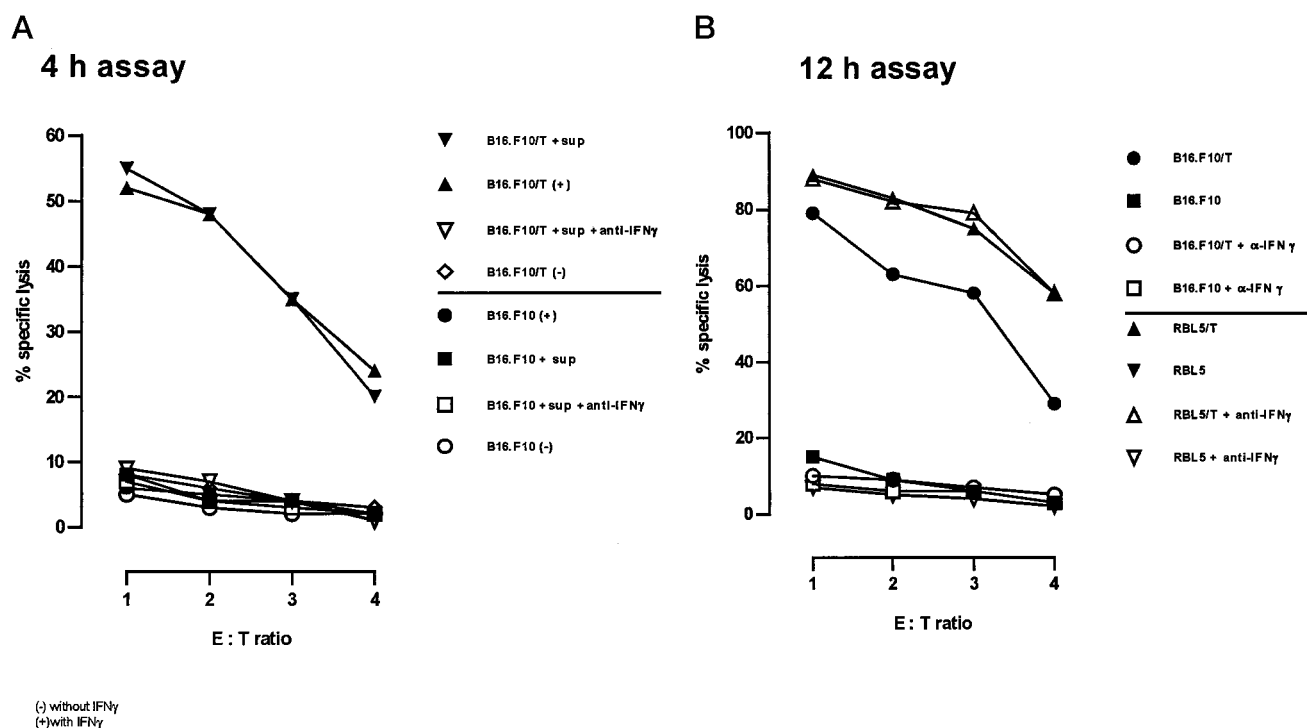
**FIGURE 5.** Specific CTL-mediated lysis of transfected B16.F10 cells. CTLL specific for HBsAg (A, C, E) or T-Ag (B, D, F) were generated in C57BL/6 mice by DNA vaccination. In a 4-h  $^{51}\text{Cr}$  release assay, these CTLL specifically lysed IFN- $\gamma$ -pretreated, but not nonpretreated, transfected B16.F10 targets (C, D). In a 12-h assay, these CTLL showed specific cytolytic reactivity toward noninduced B16.F10 transfectants (E, F). E:T ratios are 20:1, 10:1, 5:1, and 2.5:1.

subset was nonviable (stained with the nucleic acid dye 7-AAD) and apoptotic (reacted with the apoptosis marker annexin V and reacted in TUNEL stain). From the cell sorter-purified  $\text{CD95}^+\text{CD95L}^+$  subset, no cells could be expanded in vitro and no viable cells were recovered 12 to 24 h after sorting, indicating that this B16 subset is irreversibly programmed for death. Ligation of surface CD95 with Ab on  $\text{CD95}^+\text{CD95L}^+$  B16 cells may have triggered apoptosis, but this seems unlikely in view of the observation that cell sorter-purified  $\text{CD95}^+\text{CD95L}^-$  B16 cells readily grew in vitro (data not shown). Cells of the transfected B16.F10/S

and B16.F10/T sublines showed a similar surface expression pattern of CD95 and CD95L (data not shown).

#### *B16 cells stimulate IFN- $\gamma$ release by CTL and are susceptible to specific CTL-mediated lysis*

CTLL specific for MHC-I-binding epitopes of the T-Ag or the HBsAg were obtained from DNA-vaccinated C57BL/6 mice (23, 26). To detect specific IFN- $\gamma$  release of  $\text{CD8}^+$  CTLL in response to the class I-restricted presentation of epitopes of the two viral proteins by the transfected melanoma cells, CTL were cocultured



**FIGURE 6.** Specific CTL lysis of B16.F10 transfectants is IFN- $\gamma$  dependent. **A**, Medium (sup) was conditioned by coculturing  $10^6$ /ml CTLL.T cells with  $10^6$ /ml B16.F10.T cells for 12 h at 37°C. The medium was cleared after the incubation (15 min,  $2000 \times g$ ) and was used at 5% v/v; its IFN- $\gamma$  content is shown in Figure 4. The mAb R4-6A2 (anti-IFN- $\gamma$ ), which neutralizes mouse IFN- $\gamma$ , was added to some cultures at a final concentration of 20  $\mu$ g/ml; its neutralization activity is shown in Figure 4. CTLL.T lysed B16.F10/T targets treated with either sup or rIFN- $\gamma$ . Sup in which the IFN- $\gamma$  bioactivity was neutralized did not induce susceptibility to lysis in B16.F10/T targets. **B**, During a 12-h assay, B16.F10/T targets were lysed only when IFN- $\gamma$  activity was present in the cultures; neutralization of this activity suppressed the generation of a lytic interaction between CTL and the melanoma targets.

for 4 to 24 h with nontransfected B16.F10 cells or transfected B16.F10/T or B16.F10/S cells (or control RBL5, RBL5/T, or RBL5/S cells) (Fig. 4). Transfected RBL5 cells supported a specific IFN- $\gamma$  release response by CTL detectable after 4 h and 12 h; transfected B16 cells supported a specific IFN- $\gamma$  release response by CTL detectable after 12 h but not 4 h. The latter response was detectable after coculture with B16.F10/S or B16.F10/T transfectants harvested from serum-supplemented or serum-free cultures (data not shown).

In short term 4-h cytotoxic assays, CTLL did not specifically lyse B16.F10/S or B16.F10/T targets (Fig. 5, C and D) but efficiently and specifically lysed RBL5/T and RBL5/S control targets (Fig. 5, A and B). IFN- $\gamma$ -treated B16.F10/S and B16.F10/T transfectants were efficiently and specifically lysed by the respective CTLL in a 4-h assay, indicating that the up-regulated K<sup>b</sup>/D<sup>b</sup> surface expression allowed the specific cytolytic interaction of the targets with the effector cells in a short term assay (Fig. 5, C and D). When the coculture of CTL with (noninduced) B16.F10/S and B16.F10/T targets was extended to 12 h, efficient and specific CTL lysis was observed (Fig. 5, E and F).

In a 4-h assay, B16.F10 transfectants (but not nontransfected control targets) incubated for 12 h either with recombinant IFN- $\gamma$ , or with medium conditioned by CTLL (stimulated for 12 h with noninduced B16.F10 transfectants) were specifically lysed by CTL (Fig. 6A). Elimination of IFN- $\gamma$  bioactivity from medium conditioned by CTLL with the neutralizing R4-6A2 Ab suppressed the activity of these supernatants to render B16.F10 transfectants susceptible to CTL lysis in a 4-h assay. CTL-derived IFN- $\gamma$  generated during coculture of CTLL with noninduced B16.F10 transfectants hence seems to render melanoma cells susceptible to specific cytolytic attack in a 12-h assay. This was confirmed in experiments

that showed that the specific cytolysis of B16.F10 transfectants in a 12-h assay was blocked by neutralizing endogenously generated IFN- $\gamma$  bioactivity in the cultures (Fig. 6B). Similar data were obtained using B16.F10/T or B16.F10/S transfectants (data not shown).

When B16.F10/S and B16.F10/T "effector" cells were cocultured at E:T ratios of 10:1 to 1:1 with radiolabeled CTLL "target" cells in a 12-h assay, no specific lysis of the T cell targets was detectable. We thus found no evidence for specific cytolytic reactivity of the tumor cells against the immune effector cells (data not shown).

#### *Tumorigenicity of the B16 melanoma cell line and its transfected sublines in syngeneic, immune, or nonimmune hosts*

Subcutaneous transfer of  $10^3$  to  $10^6$  B16 cells into C57BL/6 mice induced aggressively growing, lethal tumors in most transplanted animals (Table I). The B16 transfectants expressing either the HBsAg or the T-Ag showed a similar aggressive growth in vivo. Only when low numbers of B16.F10/S transfectants were transplanted did a significantly reduced number of animals develop melanomas (Table I).

Growth of nontransfected B16.F10 cells and transfected B16.F10/S and B16.F10/T cells was tested in DNA-vaccinated hosts. The data in Figure 7 and Table II demonstrate that DNA vaccination specifically protected mice against the aggressive growth of a melanoma that expressed the viral Ag against which the host was immunized. MHC-I expression was readily detectable on transfected B16 cells freshly isolated from melanomas growing in immune hosts. These data confirm observations in other mouse tumor models (26, 29–32). Although MHC-I surface expression is



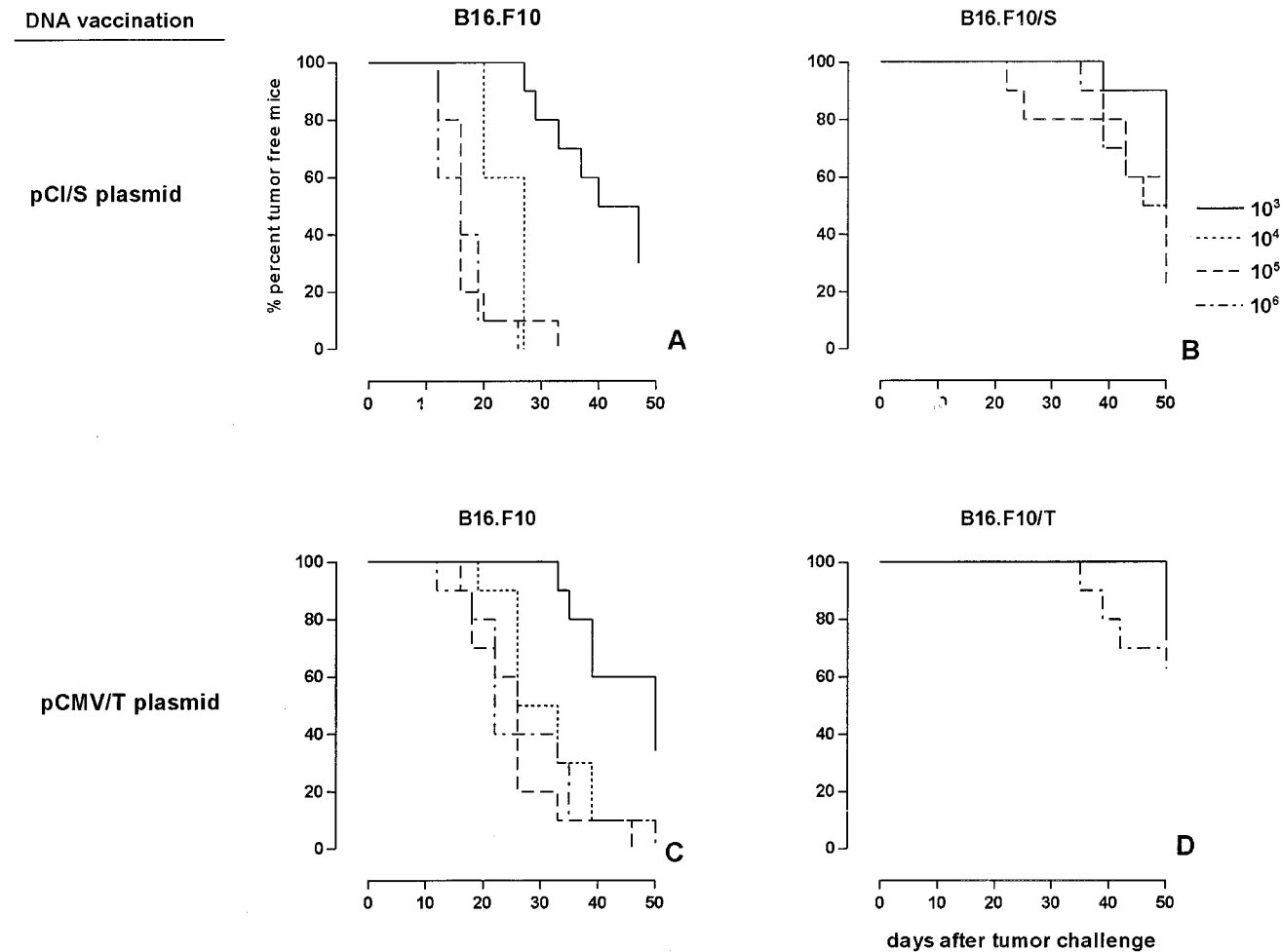
Table I. Subcutaneous growth of transfected B16 cells in syngeneic hosts

| Group | Cell Line <sup>a</sup> | No. of Cells Transferred <sup>b</sup> | Tumor Incidence <sup>c</sup> |                                 |                      | <i>p</i> <sup>d</sup> |
|-------|------------------------|---------------------------------------|------------------------------|---------------------------------|----------------------|-----------------------|
|       |                        |                                       | %                            | Tumor-bearing/transplanted mice | Comparison of groups |                       |
| 1     | B16.F10                | 10 <sup>3</sup>                       | 60                           | 6/10                            |                      |                       |
| 2     | B16.F10                | 10 <sup>4</sup>                       | 100                          | 10/10                           |                      |                       |
| 3     | B16.F10                | 10 <sup>5</sup>                       | 100                          | 10/10                           |                      |                       |
| 4     | B16.F10                | 10 <sup>6</sup>                       | 100                          | 10/10                           |                      |                       |
| 5     | B16.F10/S              | 10 <sup>3</sup>                       | 50                           | 5/10                            | 5/1                  | 1.000                 |
| 6     | B16.F10/S              | 10 <sup>4</sup>                       | 70                           | 7/10                            | 6/2                  | 0.003*                |
| 7     | B16.F10/S              | 10 <sup>5</sup>                       | 100                          | 10/10                           | 7/3                  |                       |
| 8     | B16.F10/S              | 10 <sup>6</sup>                       | 100                          | 10/10                           | 8/4                  |                       |
| 9     | B16.F10/T              | 10 <sup>3</sup>                       | 50                           | 5/10                            | 9/1                  | 1.000                 |
| 10    | B16.F10/T              | 10 <sup>4</sup>                       | 100                          | 10/10                           | 10/2                 |                       |
| 11    | B16.F10/T              | 10 <sup>5</sup>                       | 100                          | 10/10                           | 11/3                 |                       |
| 12    | B16.F10/T              | 10 <sup>6</sup>                       | 100                          | 10/10                           | 12/4                 |                       |

<sup>a</sup> Nontransfected B16 cells (B16.F10) or B16 cells transfected with HBsAg-encoding BMG/HBS plasmid DNA (B16.F10/S) or BMG/T-Ag.1 plasmid DNA (B16.F10/T) were transplanted into C57BL/6 mice.  
<sup>b</sup> 10<sup>3</sup>, 10<sup>4</sup>, 10<sup>5</sup>, or 10<sup>6</sup> cells suspended in 0.05 ml PBS were injected s.c. into the left lateral flank.  
<sup>c</sup> Tumor growth was monitored every second or third day. The percentage (%) of mice developing a tumor and the number of tumor-bearing mice/total number of transplanted mice are listed. Data from the experiment plotted in Figure 8 are shown.  
<sup>d</sup> *p* values were calculated using Fisher's exact test; \*, statistically significant.

low and MHC-II surface expression is undetectable in FCM analyses of noninduced B16 cells, tumor eradication operated efficiently in immune hosts.

Rejection of transfected B16.F10 melanomas by immune hosts required CD4<sup>+</sup> and CD8<sup>+</sup> T cells. The CD4<sup>+</sup> T cell subset, the CD8<sup>+</sup> T cell subset, or both subsets were in vivo deleted in



**FIGURE 7.** Subcutaneous growth of nontransfected and transfected B16.F10 melanoma cells in immune, syngeneic hosts. C57BL/6 mice were DNA vaccinated against HBsAg or T-Ag. These mice showed efficient, MHC-I-restricted CTL reactivity against the respective viral Ags. These mice were transplanted with 10<sup>3</sup>, 10<sup>4</sup>, 10<sup>5</sup>, or 10<sup>6</sup> nontransfected B16.F10 or transfected B16.F10/S or B16.F10/T tumor cells. The fraction of tumor-free/tumor-bearing mice was scored for 50 days posttransplantation. Mice with tumors of a diameter >1 cm were sacrificed.

Table II. Subcutaneous growth of transfected or nontransfected B16 melanoma cells in DNA-vaccinated, syngeneic hosts

| Group | Cells Transferred <sup>a</sup> |                 |                             | Tumor Incidence <sup>c</sup> |                                 |                |                       |
|-------|--------------------------------|-----------------|-----------------------------|------------------------------|---------------------------------|----------------|-----------------------|
|       | Cell line                      | No. of cells    | Mice Immune to <sup>b</sup> | %                            | Tumor-bearing/transplanted mice | Compare groups | <i>p</i> <sup>d</sup> |
| 1     | B16.F10                        | 10 <sup>3</sup> | HBsAg(S)                    | 70                           | 7/10                            |                |                       |
| 2     | B16.F10                        | 10 <sup>4</sup> | HBsAg(S)                    | 100                          | 10/10                           |                |                       |
| 3     | B16.F10                        | 10 <sup>5</sup> | HBsAg(S)                    | 100                          | 10/10                           |                |                       |
| 4     | B16.F10                        | 10 <sup>6</sup> | HBsAg(S)                    | 100                          | 10/10                           |                |                       |
| 5     | B16.F10/S                      | 10 <sup>3</sup> | HBsAg(S)                    | 10                           | 1/10                            | 5/1            | 0.019*                |
| 6     | B16.F10/S                      | 10 <sup>4</sup> | HBsAg(S)                    | 50                           | 5/10                            | 6/2            | 0.033*                |
| 7     | B16.F10/S                      | 10 <sup>5</sup> | HBsAg(S)                    | 60                           | 6/10                            | 7/3            | 0.087                 |
| 8     | B16.F10/S                      | 10 <sup>6</sup> | HBsAg(S)                    | 50                           | 5/10                            | 8/4            | 0.033*                |
| 9     | B16.F10                        | 10 <sup>3</sup> | T-Ag                        | 60                           | 6/10                            |                |                       |
| 10    | B16.F10                        | 10 <sup>4</sup> | T-Ag                        | 90                           | 9/10                            |                |                       |
| 11    | B16.F10                        | 10 <sup>5</sup> | T-Ag                        | 100                          | 10/10                           |                |                       |
| 12    | B16.F10                        | 10 <sup>6</sup> | T-Ag                        | 100                          | 10/10                           |                |                       |
| 13    | B16.F10/T                      | 10 <sup>3</sup> | T-Ag                        | 0                            | 0/10                            | 13/9           | 0.0104*               |
| 14    | B16.F10/T                      | 10 <sup>4</sup> | T-Ag                        | 0                            | 0/10                            | 14/10          | 0.0001*               |
| 15    | B16.F10/T                      | 10 <sup>5</sup> | T-Ag                        | 0                            | 0/10                            | 15/11          | <0.0001*              |
| 16    | B16.F10/T                      | 10 <sup>6</sup> | T-Ag                        | 40                           | 4/10                            | 16/12          | 0.0006*               |

<sup>a</sup> Nontransfected B16 cells (B16.F10) or B16 cells transfected with HBsAg-encoding BMG/HBS plasmid DNA (B16.F10/S) or T-Ag-encoding vector BMG/T-Ag.1 plasmid DNA (B16.F10/T) were transplanted into C57BL/6 mice; 10<sup>3</sup>, 10<sup>4</sup>, 10<sup>5</sup>, or 10<sup>6</sup> cells suspended in 0.05 ml PBS were injected s.c. into the left lateral flank.

<sup>b</sup> All mice were injected intramuscularly with 100 µg of plasmid DNA 3 wk before the s.c. tumor cell transplantation. Mice were injected with either pCI/S vector DNA (mice immune to HBsAg or S (small HBsAg)), or pCMV/T vector DNA (mice immune to T-Ag).

<sup>c</sup> Tumor growth was monitored every second or third day. The percentage (%) of mice developing a tumor and the number of tumor-bearing mice/total number of transplanted mice are listed. The data in this table and in Figure 9 are from the same experiment.

<sup>d</sup> *p* values were calculated using Fisher's exact test; \*, statistically significant.

vaccinated mice by Ab treatment before/after tumor cell transfer (Table III). Mice were monitored for 4 mo posttreatment/post-tumor cell transfer for the appearance of tumors. The (almost complete) elimination of T cells from the immune host abolished its capacity to specifically resist tumor growth, while the selective elimination of only CD4<sup>+</sup> T cells or only CD8<sup>+</sup> T cells did not. Rejection in this system is hence T dependent, and requires both CD4<sup>+</sup> and CD8<sup>+</sup> T cell subsets. Similar observations were made using B16.F10/T or B16.F10/S transfectants.

T cells were prepared from mice primed, 3 wk previously, by DNA vaccination to either T-Ag or HBsAg. The cells were adoptively transferred into syngeneic hosts bearing a melanoma that had reached a diameter of 3 to 6 mm (derived from transfected, T-Ag-expressing B16.F10 cells). Transfer of T-Ag-specific but not HBsAg-specific T cell populations led to regression of s.c. growing melanomas. The histopathology of regressing tumors indicated that a large fraction of the tumor cells underwent apoptosis detectable by the TdT-mediated in situ TUNEL technique (Fig. 8). Apoptotic cells were located predominantly at the border to the adjacent stromal tissue, whereas few apoptotic events were detectable in the central areas of tumor nodules. The fraction of apoptotic cells was >10-fold higher in tumor-bearing mice transplanted with T cells specifically immune to the viral Ag that the tumor cells expressed (i.e., the SV40 T-Ag) than in mice transplanted with T cells primed to an irrelevant control viral nucleoprotein. In vivo, rejection of melanomas by primed T cells is thus accompanied by apoptosis of the tumor cells. Injection of 100 U of mouse rIFN-γ into growing B16.F10 melanomas in vivo did not yield a similar picture.

## Discussion

Stable B16.F10 melanoma transfectants expressing either the HBsAg protein or the T-Ag nucleoprotein were established. In H-2<sup>b</sup> mice, HBsAg is a "weak" immunogen for class I-restricted CTL, and T-Ag is a "strong" immunogen for this T cell subset (23,

26). Transplantation of genetically modified tumor cells that stably express well defined, CTL-stimulating Ags has been used in many experimental systems to mimic immune responses to tumor-associated Ags. The efficacy of tumor-rejecting CTL responses have been studied with B16 cells transfected with Ag-encoding expression plasmids (32–34). In our system, the parental B16.F10 line and its transfected sublines showed a similar tumorigenicity in vivo in syngeneic C57BL/6 hosts. Only after transfer of 10<sup>4</sup> B16.F10/S cells/mouse did we observe a statistically significant, lower incidence of tumor development (Table I); we have published similar data on the P815 mastocytoma system (29).

DNA vaccination efficiently primes HBsAg- or T-Ag-specific, class I-restricted CTL responses to HBsAg and to T-Ag in H-2<sup>b</sup>

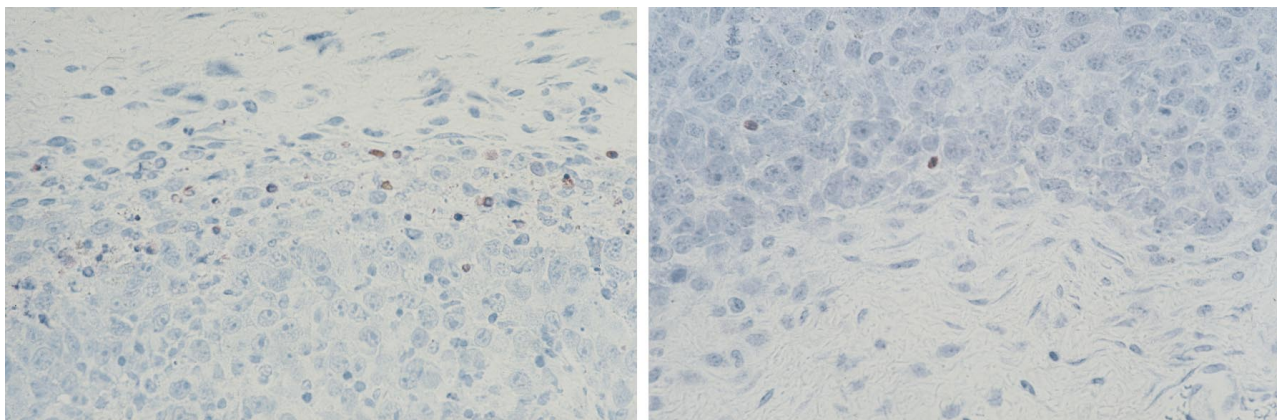
Table III. T-dependent rejection of transfected B16.F10 melanoma cells in immune hosts

| Group | Vector Used for Vaccination <sup>a</sup> | Treated with <sup>b</sup>                           | Number of Rejecting B16.F10/S-Transplanted Mice <sup>c</sup> |
|-------|------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------|
| 1     | pCI/S                                    | mAb YTS 191.1 (anti-CD4)                            | 5/5                                                          |
| 2     | pCI/S                                    | mAb YTS 169.4 (anti-CD8)                            | 5/5                                                          |
| 3     | pCI/S                                    | mAb YTS 191.1 (anti-CD4) + mAb YTS 169.4 (anti-CD8) | 0/5                                                          |
| 4     | pCI/S                                    | Nontreated                                          | 5/5                                                          |
| 5     | None                                     | Nontreated                                          | 0/5                                                          |

<sup>a</sup> Mice were vaccinated intramuscularly with 100 µg pCI/S plasmid DNA (encoding HBsAg) 3 wk before tumor cell transplantation.

<sup>b</sup> CD4<sup>+</sup> or CD8<sup>+</sup> T cell subsets were deleted in vivo by repeated injections of anti-CD4 mAb YTS 191.1, anti-CD8 mAb YTS 169.4, or both mAbs (as described in *Materials and Methods*). Stable deletion of the respective T cell subset was demonstrated in all treated mice.

<sup>c</sup> Mice were transplanted s.c. with 10<sup>5</sup> transfected B16.F10/S melanoma cells; mice were monitored for tumor growth for 4 mo posttransplantation. The number of rejecting/number of transplanted mice is shown.



**FIGURE 8.** Extensive apoptosis is evident during T-dependent regression of B16 melanomas. Splenic (B cell-depleted) T cells primed to T-Ag (*left panel*) or to HbS-Ag (*right panel*) were adoptively transferred into B16.F10/T tumor-bearing mice. Tumors were removed 2 days after T cell transfer and fixed in formalin. Apoptosis was detected using the TUNEL technique. Numerous apoptotic melanoma cells are visible, predominantly at the margin of the tumor (*left*). In contrast, only a few apoptotic bodies were detected after adoptive transfer of T cells primed to the irrelevant Ag HBsAg (*right*). Each photomicrograph is representative of a total of four tumors analyzed. Original magnification,  $\times 400$ .

mice (23, 26), but transfer of B16.F10/S or B16.F10/T cells into naive C57BL/6 mice did not (data not shown). This observation at least partly explains the aggressive *in vivo* tumor growth of transfected B16.F10/S or B16.F10/T cells. The low immunogenicity of nontransfected or transfected B16.F10 melanoma cells may be due to their deficient costimulator and cytokine expression profile, as well as to their inefficient MHC-restricted epitope presentation. The B16.F10 line used showed no (or very low) expression of the costimulator molecules CD48 (the ligand for CD2), CD54/ICAM-1 (binding LFA-1), CD80/B7-1, and CD86/B7-2 (binding CD28 or CTLA4). Expression of transfected B7-1/CD80 and/or B7-2/CD86 genes in B16 melanoma cells has been reported to induce anti-tumor immunity and to have an antimetastatic effect (35–37). It has been reported that B16 cells express IL-2 and the IL-2 receptors (38), but we found no IL-2 bioactivity in medium conditioned by the B16.F10 cells that we used (data not shown). Surface expression of MHC-I and MHC-II molecules by B16.F10 cells was low, but was strikingly up-regulated by IFN- $\gamma$  treatment. This melanoma cell line is thus inefficient in the immunogenic, MHC-restricted epitope presentation to T cell precursors.

Although inefficient in priming a T cell response, B16 melanoma cells are nevertheless targets for various immune effector cells. *In vivo* growth of B16 melanomas can be eradicated by natural cell-mediated cytotoxicity (39), activated macrophages (40, 41), or CD8<sup>+</sup> CTL (1, 37, 40, 42). Although CD8<sup>+</sup> CTL seem to be involved in the rejection of B16 melanomas, CD4<sup>+</sup> T cells, and possibly other immune effector mechanisms, seem to contribute to resistance to this tumor because CD8 T cell-depleted, immune mice showed resistance to this tumor. Rejection of B16 melanomas has been reported to be facilitated by the cytokines IL-1 $\beta$  (43, 44), IL-2 (45–48), IL-4 (49), IL-10 (50, 51), IL-12 (52, 53), granulocyte-macrophage (GM)-CSF (36, 54–57), monocyte (M)-CSF (52), TNF- $\alpha$  (43), and IFN- $\gamma$  (40, 41, 58), but inhibited by TGF- $\beta$  (45, 59). IFN- $\gamma$  treatment *in vivo* and *in vitro* renders B16 melanoma cells susceptible to lysis by activated macrophages (1, 41, 60) and by CTL (this paper). Our data indicate that the interaction of Ag-presenting B16 cells with MHC-I-restricted CTL involved multiple events. 1) Specific MHC-I-restricted recognition of the viral epitopes presented by noninduced B16 cells triggers IFN- $\gamma$  release by CTL but not their cytolytic program. 2) IFN- $\gamma$  stimulation of B16 cells up-regulates MHC-I-restricted epitope presentation and allows the specific, cytolytic interaction between

effector and target cell. 3) IFN- $\gamma$  stimulation enhances the CD95/CD95L suicide program of a subset of B16 melanoma cells.

CTL that specifically recognize epitopes of the viral proteins HBsAg or T-Ag, in the context of MHC-I molecules presented by noninduced B16 cells, release IFN- $\gamma$ . In contrast, the cytolytic program of these CTL (detectable in a 4-h <sup>51</sup>Cr release assay) was not triggered by this specific interaction. This was observed using five independently derived CTLL (data not shown). This partial T cell activation may result from low MHC-I expression and/or low antigenicity of the recognized peptide epitope. The phenomenon resembles the selective signaling model observed in altered peptide ligand-induced partial T cell activation (61). IFN- $\gamma$  produced by CTL in the immediate vicinity of B16 cells stimulates up-regulation of MHC-I- and MHC-II-restricted epitope presentation. This apparently facilitates the specific and efficient cytolysis of melanoma cells by primed CTL. In addition, IFN- $\gamma$  may also enhance other tumor-rejecting effector mechanism (e.g., macrophages, dendritic cells).

The CTL-mediated lysis of melanoma cells may not be the only or the decisive mechanism of tumor eradication. A subset of nontransfected and transfected B16.F10 cells showed surface expression of CD95 and CD95L. Almost 90% of the B16.F10 cells expressed readily detectable levels of CD95 on the surface after a 12-h stimulation by 30 IU/ml murine IFN- $\gamma$ . Furthermore, a well-defined 30 to 40% subset of the IFN- $\gamma$ -stimulated B16.F10 cells coexpressed CD95L and CD95 on the cell surface; this was demonstrated by RT-PCR and by FCM analyses. IFN- $\gamma$ -stimulated B16 cells that coexpressed CD95 and CD95L were apoptotic. They were stained for the apoptosis markers annexin, by the nucleic acid dye 7-AAD, and by the TUNEL technique; and they could not be maintained in a viable state *in vitro*. In unstimulated B16.F10 cells growing under serum-free conditions, a fraction of 10 to 20% are continuously eliminated by suicide. The apoptotic fraction of these melanoma cells is higher when cells are cultured in serum (FCS)-supplemented medium. We did not detect a significant fraction of cells dying of fratricide because apoptotic CD95<sup>+</sup>CD95L<sup>−</sup> melanoma cells were always below the detection threshold. It is surprising that a tumor cell line adapted to long term culture for many years continuously and “spontaneously” generates an apoptotic subset of progeny. This may represent a “loss by differentiation” event. Analyses using maturation markers of melanocytes may clarify this point. The important point for the design of tumor



vaccines is that CTL-derived cytokines can greatly amplify this differentiation program of melanomas to an extent that might reach therapeutic value.

In vivo, specific rejection of transfected B16.F10 melanoma cells was T-dependent; and adoptively transferred, primed T cells induced massive apoptosis and regression in a s.c. growing melanoma. It is not clear to what extent this in vivo apoptosis results from direct CTL attack or from CTL-facilitated suicide. It seems of interest to test whether focusing CTL with a desired profile of cytokines to growing tumor cells is as important for tumor eradication as a direct cytolytic attack of tumor cells by these immune effector cells.

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